

Supply and Demand

Mbak 9512: Economics

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Brief Review

Equation

- A standard Tomato Sauce Pizza is \$5
 - ▶ If you add toppings, it costs \$2.5 per each topping.
 - ▶ Can you set the equation? You might use ***P*** to represent price of pizza and ***T*** to represent the topping.

$$P = 5 + 2.5T$$

- If you want to have three toppings, the price of pizza is

$$\begin{aligned} P &= 5 + 2.5(3) \\ &= 12.5 \end{aligned}$$

Brief Review

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Brief Review

Graphing an Equation

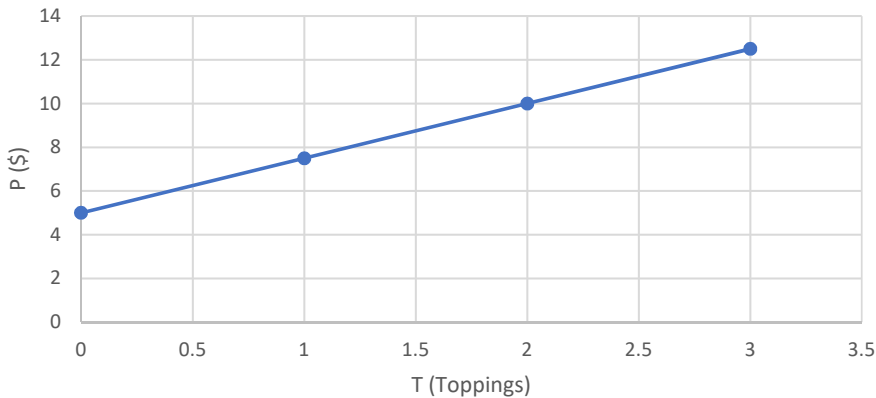
T	P
0	5
1	7.5
2	10
3	12.5
$\vdots$	$\vdots$

- We call T as an *independent variable* and P as a *dependent variable*

Brief Review

Graphing an Equation

Pizza Price

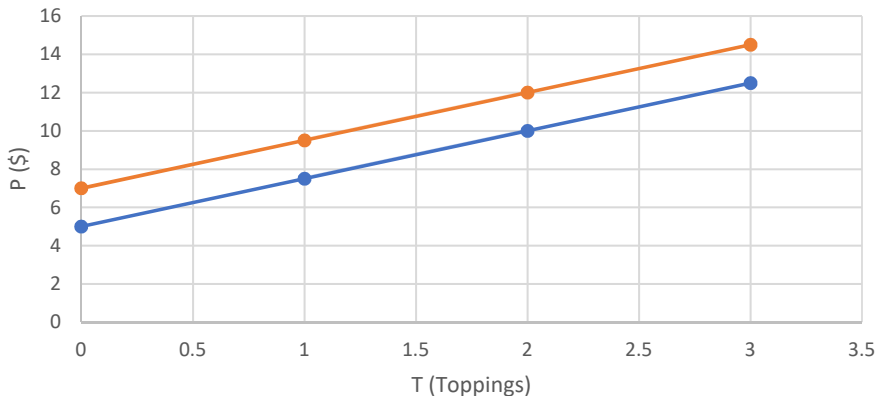


Brief Review

Intercept Change

- A standard Tomato Sauce Pizza is now \$7
- Accordingly, the equation is $P = 7 + 2.5T$

Pizza Price

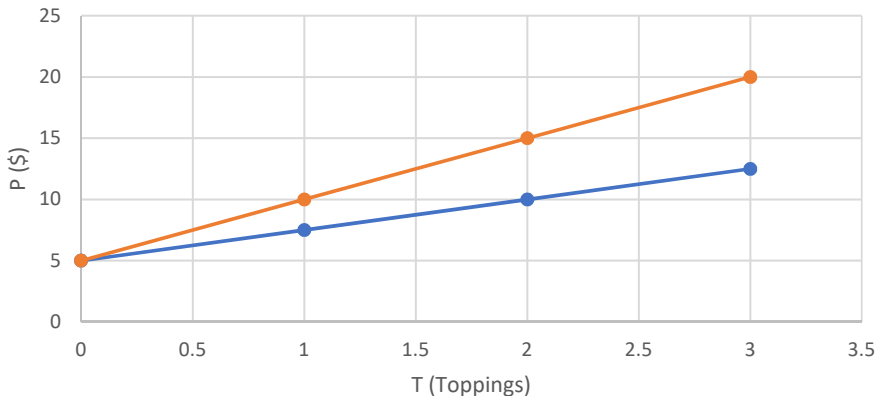


Brief Review

Slope Change

- A standard Tomato Sauce Pizza is \$5 but topping is \$5 now
- Accordingly, the equation is $P = 5 + 5T$

Pizza Price

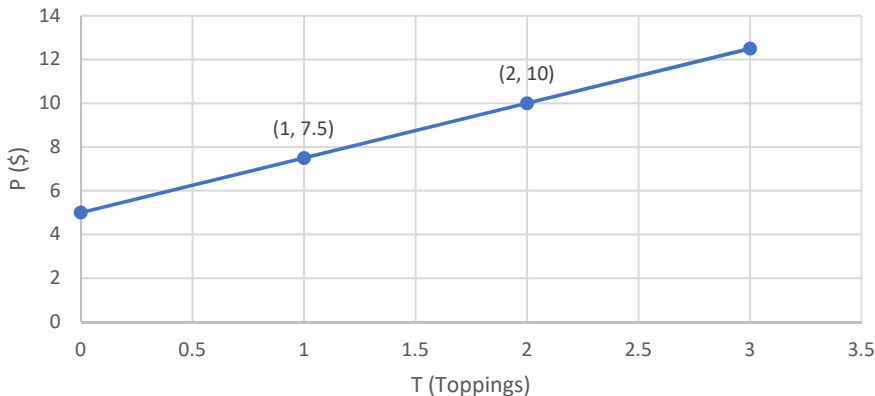


Brief Review

Deriving the Equation

- Now we have only two pairs of points
- Can you derive the equation?

Pizza Price



Brief Review

Deriving the Equation

- As it is a straight line, we can use the generic expression $P = a + bT$
- Both two points satisfy the equation above, so we have two equations and two unknown constants

$$10 = a + b \cdot 2 \quad (1)$$

$$7.5 = a + b \cdot 1 \quad (2)$$

$$\begin{aligned}(10 - 7.5) &= b \\ &= 2.5\end{aligned}$$

- then we can find a by substituting $b = 2.5$ to equation (1): $a = 5$

Brief Review

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Readings

- Chapter 4. The Market Forces of Supply and Demand

- ▶ Markets and Competition
- ▶ Demand
- ▶ Supply
- ▶ Equilibrium and Excess Supply and Demand

Demand

- Why buy more or less ice-cream in a given week?
- Quantity Demanded: the amount of a good or service that a consumer is willing and able to purchase at a given price.

Demand Schedule

- A table that shows the relationship between the price of a product and the quantity of the product demanded.
- For example, let's think about Catherine's demand

P	Q_c
\$0.0	
\$0.5	
\$1.0	
\$1.5	
\$2.0	
⋮	

Demand Schedule

- A table that shows the relationship between the price of a product and the quantity of the product demanded.
- For example, let's think about Catherine's demand

P	Q_c
\$0.0	12
\$0.5	10
\$1.0	8
\$1.5	6
\$2.0	4
$\vdots$	$\vdots$

Demand Schedule

- Similarly consider Nicholas' demand

P	Q_n
\$0.0	7
\$0.5	6
\$1.0	5
\$1.5	4
\$2.0	3
$\vdots$	$\vdots$

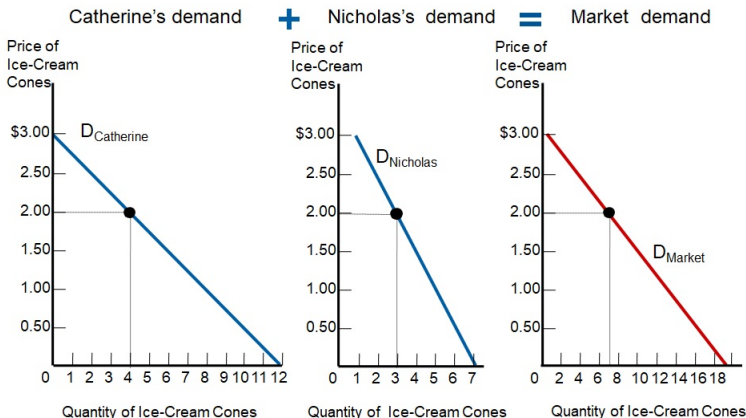
Market Demand Schedule

- The Sum of Individual Demands

Catherine		Nicholas		Market	
P	Q_c	P	Q_n	P	Q_D
\$0.0	12	\$0.0	7	\$0.0	19
\$0.5	10	\$0.5	6	\$0.5	16
\$1.0	8	\$1.0	5	\$1.0	13
\$1.5	6	\$1.5	4	\$1.5	10
\$2.0	4	\$2.0	3	\$2.0	7
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Demand Curve

- Demand Curve: Graph depicting the relationship between the quantity demanded and the price
- Market Demand Curve: Add up the quantity demanded for all consumers of ice-cream at various price levels.

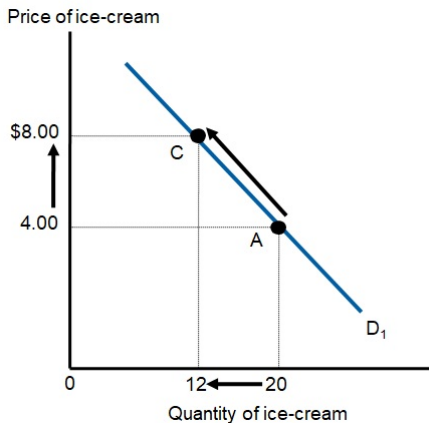


Demand Curve

- As price falls, the quantity demanded rises because:
 - ▶ substitution effect of a price change (cheese cake becomes less attractive)
 - ▶ income effect of a price change (increases purchasing power)

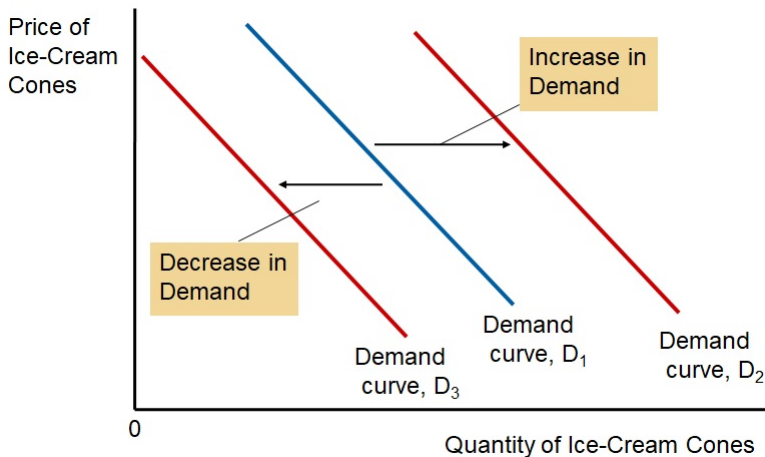
Change in the Quantity Demanded

- The change in the quantity that people wish to buy that occurs **in response to** a change in price.
- change in $P \Rightarrow$ change in Q_D



Change in Demand

- At every price, the quantity demanded is higher/lower than before
- Shift of the entire demand curve



Change in Demand

What shifts demand curve

- Income
 - ▶ got fired, promotion
- Prices of related goods
 - ▶ **Complements:** Goods and services that are used together
 - ★ ice-cream and waffle, Chips and avocado dip
 - ▶ **Substitutes:** Goods and services that can be used for the same purpose
 - ★ ice-cream vs frozen yogurt, SUV vs Hybrid Car
- Change in tastes
- Expectations about the future
 - ▶ Expect an increase in income
 - ▶ Expect higher prices
- Number of buyers

Change in Demand

Summary Table

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

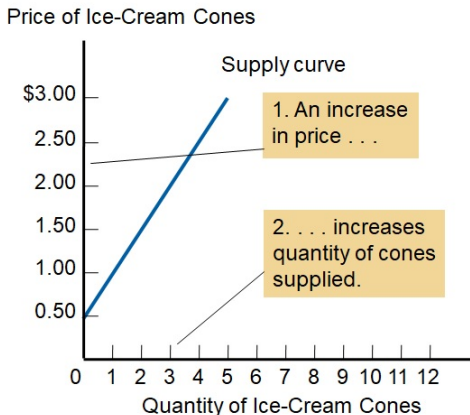
Supply Schedule

- A table that shows the relationship between the price of a product and the quantity of the product supplied.
- Consider Ben's supply schedule

P	Q_b
\$0.0	0
\$0.5	0
\$1.0	1
\$1.5	2
\$2.0	3
$\vdots$	$\vdots$

Supply Curve

- Supply Curve: Graph depicting the relationship between the quantity supplied and the price



- As price **rises**, the quantity supplied **rises** because:
 - the profit to suppliers rises relative to their production costs

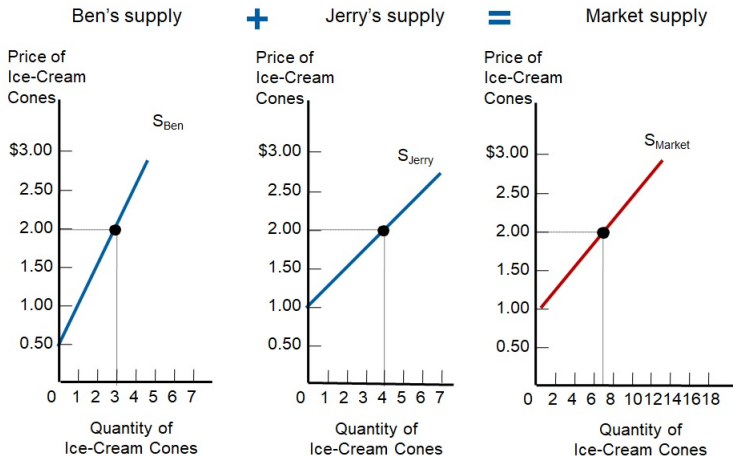
Market Supply Schedule

- The Sum of Individual Supply

Ben		Jerry		Market	
P	Q_b	P	Q_j	P	Q_s
\$0.0	0	\$0.0	0	\$0.0	0
\$0.5	0	\$0.5	0	\$0.5	0
\$1.0	1	\$1.0	0	\$1.0	1
\$1.5	2	\$1.5	2	\$1.5	4
\$2.0	3	\$2.0	4	\$2.0	7
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

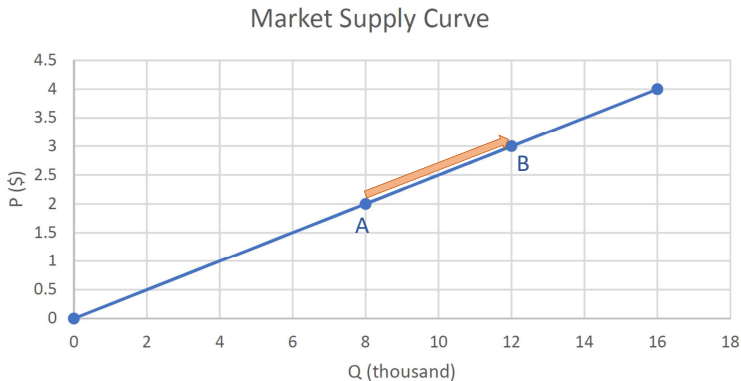
Market Supply Curve

- Market Supply Curve: Add up the quantity supplied horizontally for all suppliers of ice-cream at various price levels.



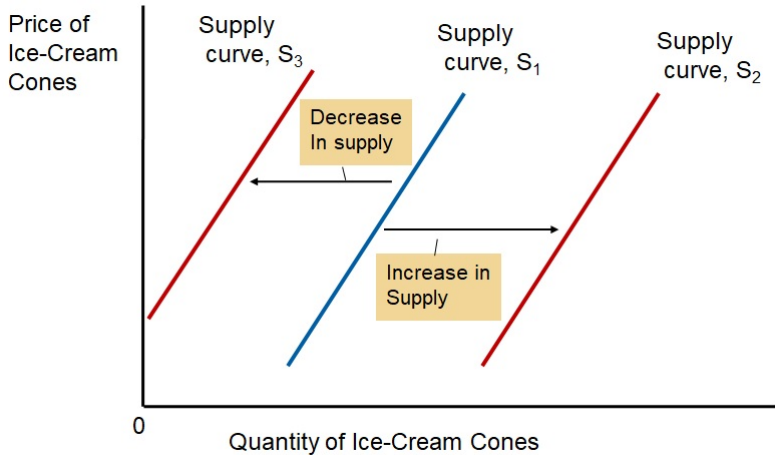
Change in the Quantity Supplied

- change in $P \Rightarrow$ change in Q_S



Change in Supply

- At every price, the quantity supplied is higher/lower than before
- Shift of the entire supply curve



Change in Supply

What shifts demand curve

- Input prices
 - ▶ milk, sugar
- Technology
 - ▶ new ice-cream machine which makes ice-cream much faster
- Expectations about future
 - ▶ Expected higher prices
- Number of sellers in ice-cream market

Change in Supply

Summary Table

Variable	A Change in This Variable . . .
Price of the good itself	Represents a movement along the supply curve
Input prices	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

Market Equilibrium

- The interaction of suppliers/producers and consumers/buyers in a market will determine the price of a good and the quantity that is sold.

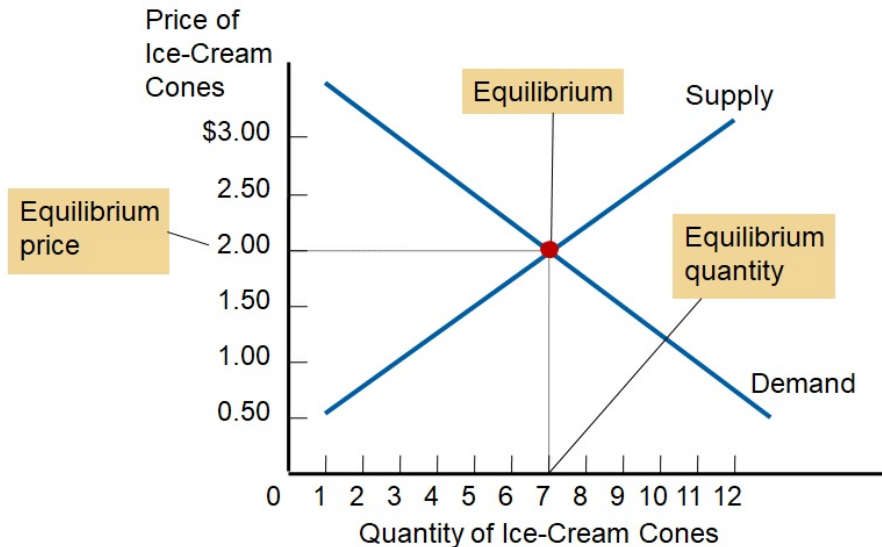
P	Q_D	Q_S
\$0.0	19	0
\$0.5	16	0
\$1.0	13	1
\$1.5	10	4
\$2.0	7	7
\$2.5	4	10
\$3.0	1	13
$\vdots$	$\vdots$	$\vdots$

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$\vdots$	$\vdots$	$\vdots$

Market Equilibrium

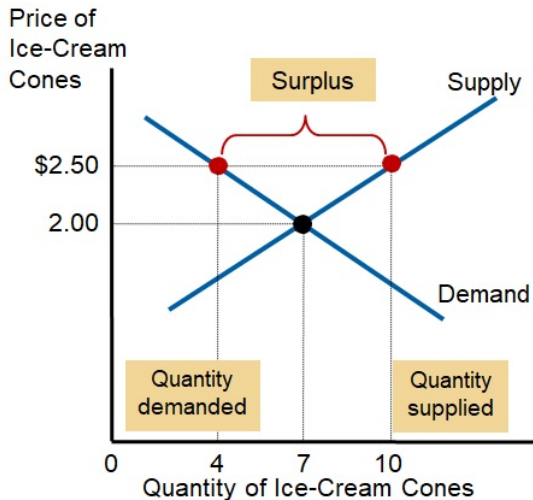


Market Equilibrium

- **Exogenous variable:** a factor whose value is determined by factors or variables outside the causal system
 - ▶ sugar price, milk price, frozen custard price
- **Endogenous variable:** a factor whose value is determined by factors or variables in the causal system
 - ▶ price of ice-cream and quantity of ice-cream

Market Equilibrium

- This equilibrium is stable because:
 - ▶ At prices **above** the equilibrium price, there is *excess supply* (the quantity supplied > the quantity demanded) $\Rightarrow$ prices **fall**



Market Equilibrium

- This equilibrium is stable because:
 - ▶ At prices **below** the equilibrium price, there is *excess demand* (the quantity supplied < the quantity demanded) $\Rightarrow$ prices **rise**

